

Addressing the risks associated with the global semiconductor supply chain

Leveraging Data and AI Governance

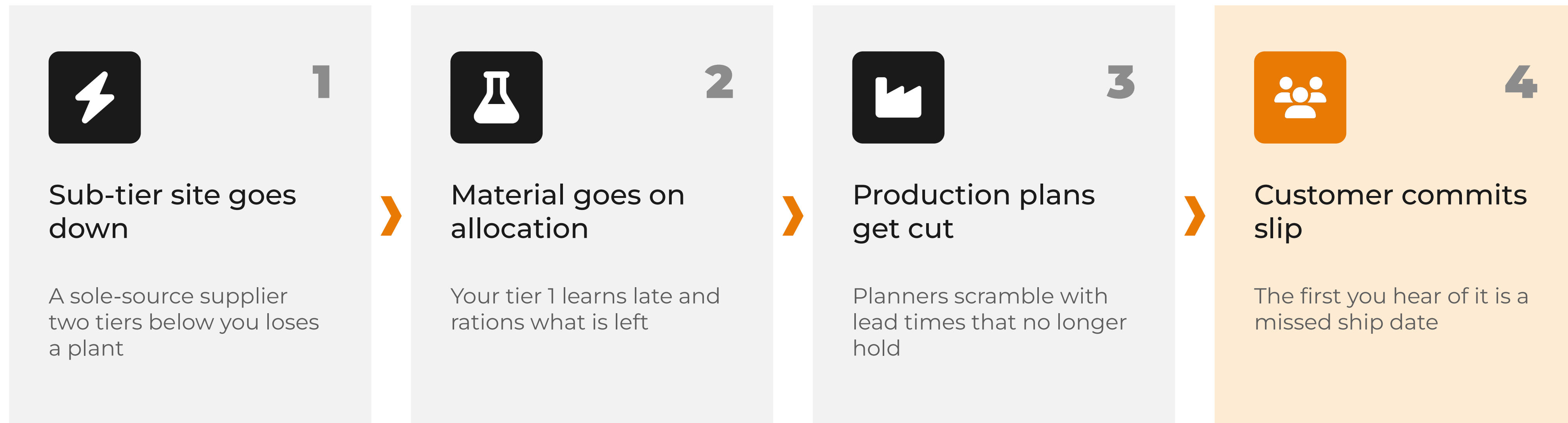
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ILLUSTRATIVE SCENARIO

One missing input can stall output downstream



The failure starts where most of us have the least visibility.

The risk map

Where the global chain is fragile, and why it matters now

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THE RISK MAP

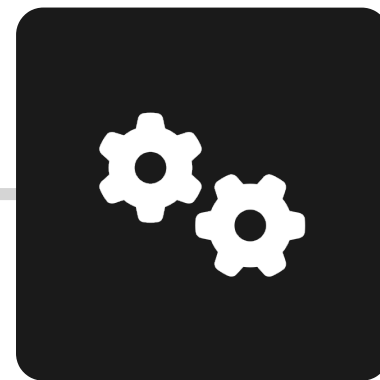
Specialization made the chain efficient and fragile



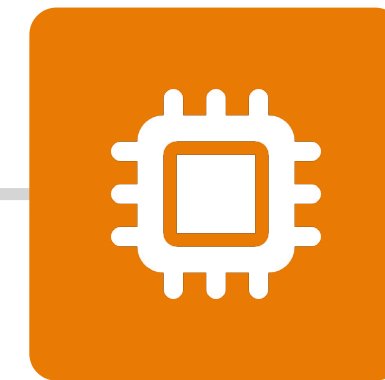
Design & IP



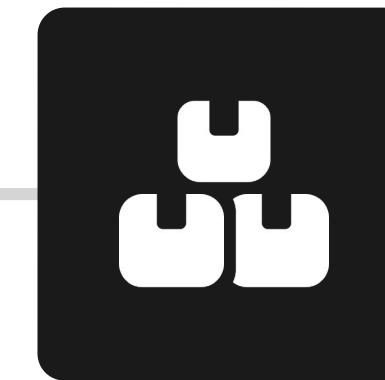
Materials & gases



Equipment



Wafer fab



Assembly & test



Customers

Each step is specialized

Few qualified sources exist for many inputs, and qualifying a new one takes months to years.

Each hand-off can cross a border

Parts and materials often move through several countries before a finished chip ships, so trade policy, logistics, and local events all add risk.

THE RISK MAP

Three pressures are tightening the chain right now

100%

Advanced logic in two places

In 2022, all sub-10 nm logic capacity was in Taiwan (69%) and South Korea (31%). A regional conflict or blockade would hit nearly every advanced product.

Source: SIA & BCG (2024)

99%

Gallium from one country

China produced 99% of the world's primary gallium in 2024, and has put export licensing and bans on gallium and germanium since 2023.

Source: USGS Mineral Commodity Summaries 2025

HBM

AI demand absorbs capacity

Demand for high-bandwidth memory and AI hardware has stretched lead times and strained sub-10 nm fab capacity.

Source: Altium

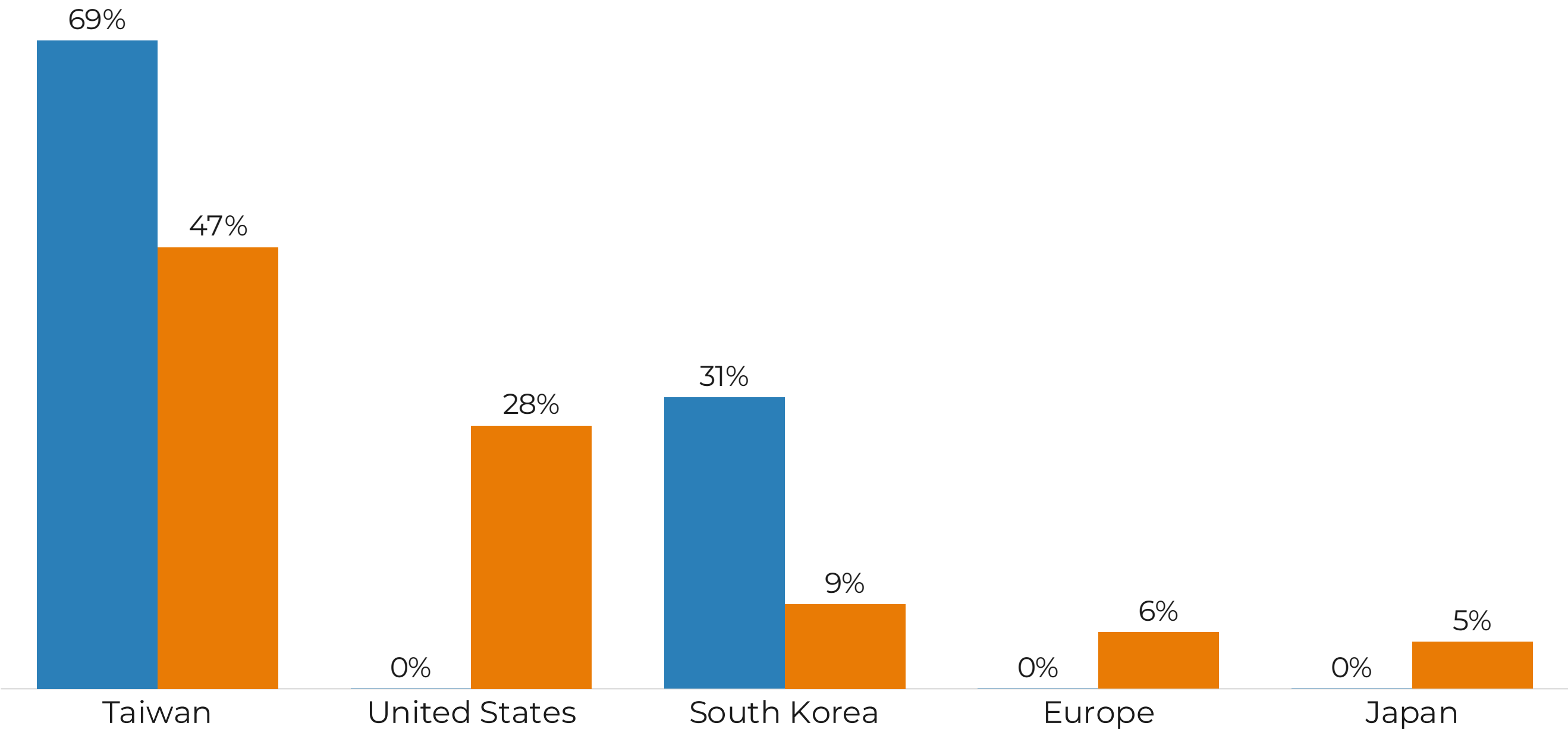
Each one lands on a node most companies can't see well today.

THE RISK MAP

Advanced logic stays concentrated as new fabs arrive

Share of global sub-10 nm logic wafer capacity

■ 2022 ■ 2032 (projected)



2 places

held all sub-10 nm logic capacity in 2022

47%

Taiwan's projected share in 2032, still the largest

28%

projected U.S. share by 2032, up from 0%

Source: SIA & BCG, Emerging Resilience in the Semiconductor Supply Chain (May 2024). 2032 values are projections; the remaining ~5% in 2032 includes China and other regions.

THE RISK MAP

Six kinds of risk tend to hit the same few nodes



Concentration

Sole- and single-source inputs, and sites that serve many products at once



Geopolitics & trade

Export controls, entity lists, tariffs, and shifting rules of origin



Natural hazards & utilities

Earthquakes, storms, drought, and power or water outages at key sites



Logistics

Port, air freight, and customs delays for time-critical materials



Demand swings

Forecast whiplash that turns into over-ordering, then shortages



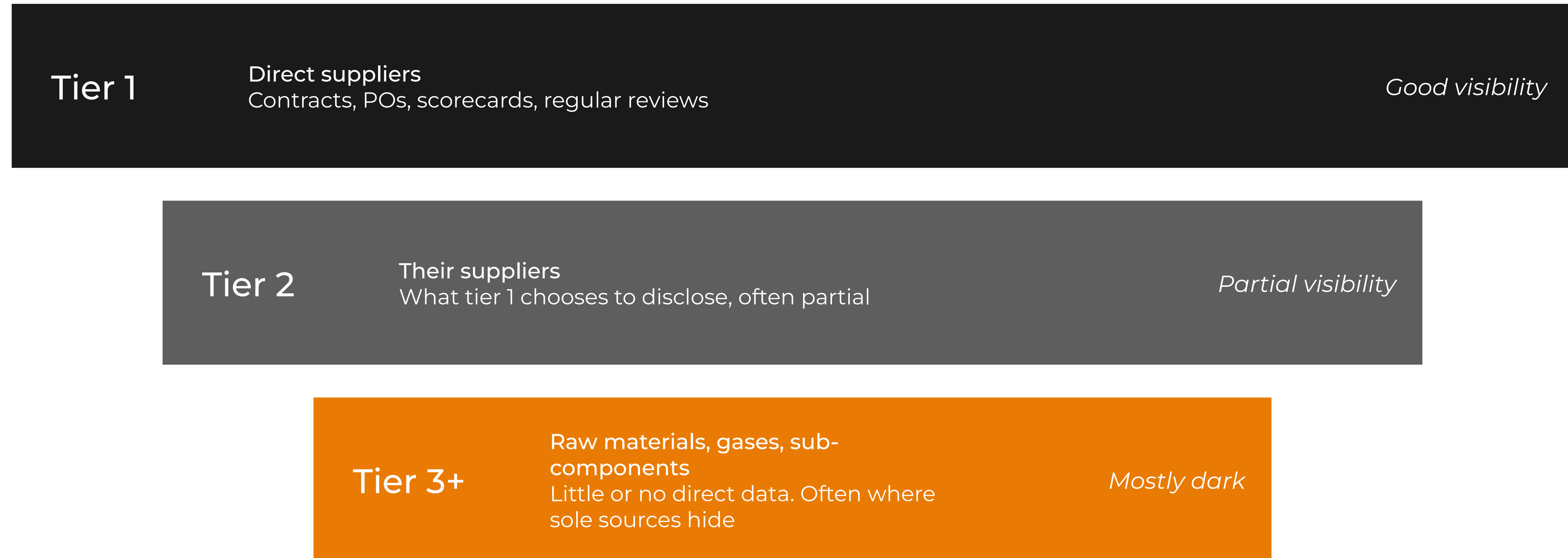
Cyber & IP

Attacks on suppliers and leaks of sensitive design or process data

When several of these overlap at one node, that node is where to look first.

THE RISK MAP

Most exposure sits below tier 1, where visibility drops off



You can't govern data you never collect. The first job is deciding which sub-tier nodes you need to see.

Why data is the gap

The data exists. Trusting and connecting it is the problem.

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WHY DATA IS THE GAP

Risk data exists, but it's scattered across systems



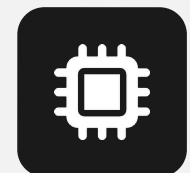
ERP / MRP



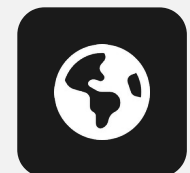
Supplier portals



Logistics feeds



MES & quality



News & trade data



Different IDs for the same thing

Part numbers, supplier codes, and site names don't match across systems.

No lineage

Nobody can say where a lead time or risk score came from, or when it was last true.

Stale master data

Planned lead times and approved sources drift from reality between reviews.

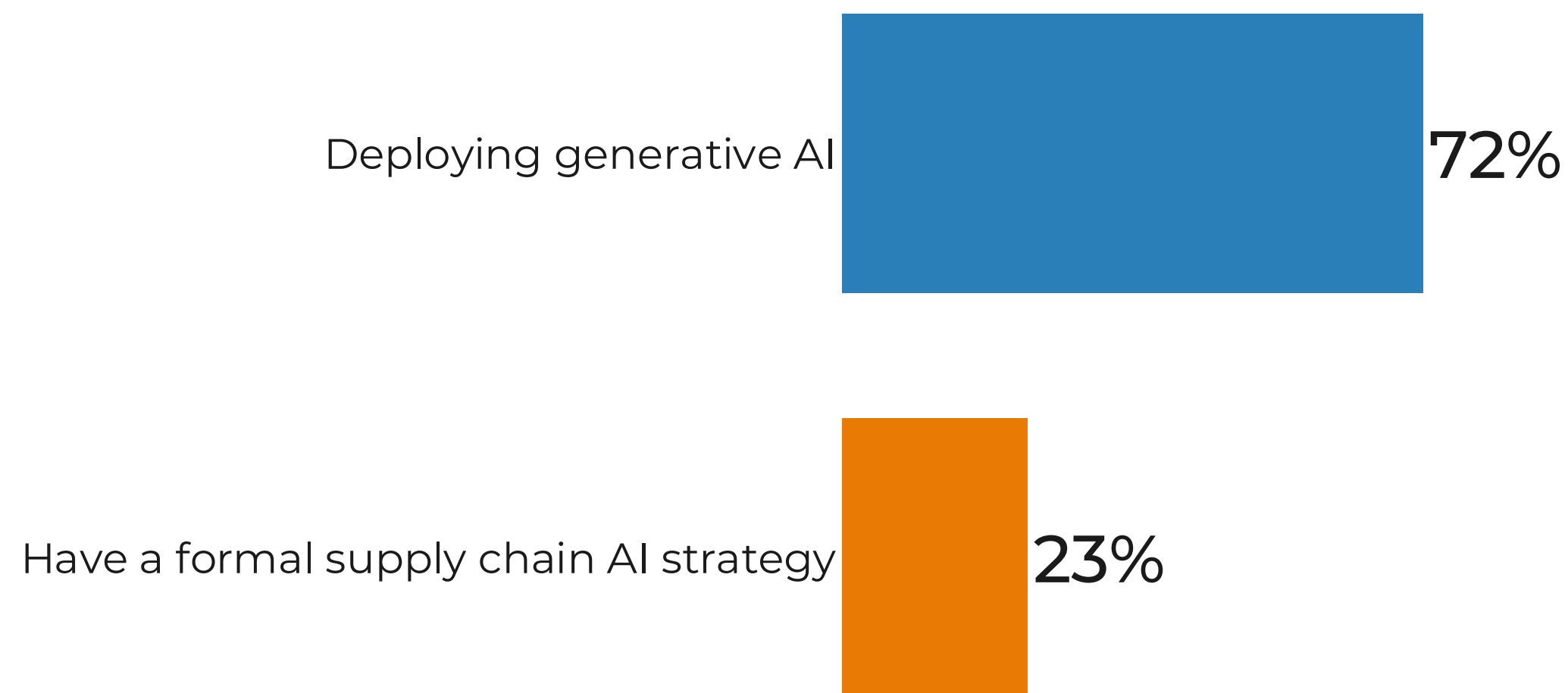
No terms for sharing

Partners hold back data because use, access, and retention were never agreed.

WHY DATA IS THE GAP

AI adoption is running ahead of AI governance

Share of supply chain organizations (Gartner surveys, 2025)



Sources: Gartner press releases, Feb 5, 2025 (GenAI deployment) and Jun 11, 2025 (AI strategy; 120 leaders surveyed). Separate surveys.

When it's done well

15% better logistics costs

35% better inventory levels

65% better service levels

Early adopters of AI-enabled supply chain management vs. slower competitors. Source: McKinsey, "Succeeding in the AI supply-chain revolution" (2021).

The gap between those two bars is where ungoverned AI risk lives.

Governance as the lever

Controls that turn risk data into decisions you can defend

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GOVERNANCE AS THE LEVER

Each type of AI model needs its own governance control



Time-series forecasting

Gradient boosting, LSTM, transformers

Predicts demand, lead times, and allocation risk

Control: Backtest on past disruptions; monitor drift

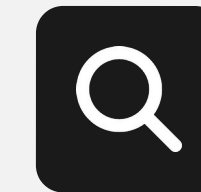


Graph models

Knowledge graphs, GNNs

Maps multi-tier BOMs to find single points of failure

Control: Lineage and a confidence level on every link



NLP & LLMs

Classifiers, language models, retrieval

Reads news, supplier notices, and regulations for early signals

Control: Ground in approved sources; log prompts and outputs

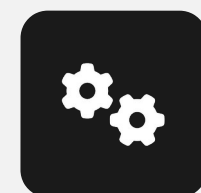


Anomaly detection

Isolation forest, autoencoders

Flags unusual shipments, quality drift, or supplier behavior

Control: Owners tune thresholds; review false alarms

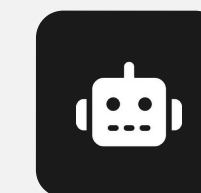


Optimization & digital twins

Solvers, simulation

Tests allocation and alternate-source scenarios

Control: Validate assumptions; a person approves the plan



Agentic AI

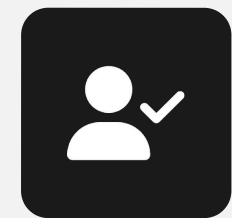
LLM agents with tools

Chains steps, such as drafting supplier queries or POs

Control: Least-privilege tools; sign-off before external actions

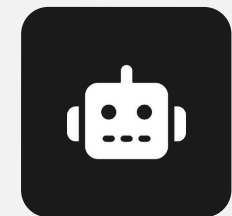
GOVERNANCE AS THE LEVER

Four layers turn risk data into decisions you can defend



Human oversight

Decision rights, escalation paths, and a named owner for every action the data or model informs



AI model risk

Validation against known events, drift monitoring, explainable outputs, and logged prompts for generative tools



Access & sharing

Least-privilege access, data-use agreements with suppliers, and audit trails on sensitive maps



Data foundation

Common part and supplier IDs, lineage, quality checks, and an owner for each critical record

Build from the bottom up. Each layer depends on the one below it.

GOVERNANCE AS THE LEVER

Map each control to the risk it reduces, and measure it

Risk	Governance control	Evidence it works
Single points of failure	AI-assisted multi-tier BOM mapping down to raw materials, with an owner per critical part	Share of critical parts traced to raw material
Export controls and sanctions	Automated screening of items, parties, and end uses against current control lists	Hours from rule change to exposure report
Allocation crunch (HBM, advanced nodes)	ML monitoring of lead-time and capacity signals at known choke points	Weeks of warning before allocation hits
AI misranks a risk	Validation on past disruptions, drift monitoring	Planner override rate, and why
Suppliers won't share	Data-use agreements, least-privilege access, audit logs	Critical suppliers sharing sub-tier data

GOVERNANCE AS THE LEVER

Use an established framework rather than inventing one

NIST AI Risk Management Framework: four functions



Govern

Policies, roles, and accountability across the AI lifecycle



Map

Context: where the model is used and what it could affect



Measure

Test, evaluate, and track the risks you mapped



Manage

Prioritize and act on those risks, and keep monitoring



ISO/IEC 42001

A certifiable AI management system standard. Useful when customers or regulators ask for proof, not just a policy.

Common pattern: design the practice with NIST, certify it with ISO. Pick one and apply it to a real use case first.

EXAMPLE | ILLUSTRATIVE, REPLACE WITH YOUR CASE

Governed data turns a scramble into a same-day decision



Step 1

Detect

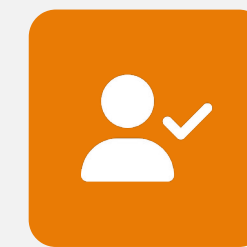
An external signal flags an outage at a specialty gas supplier. The model scores it high because the site is a sole source.



Step 2

Assess

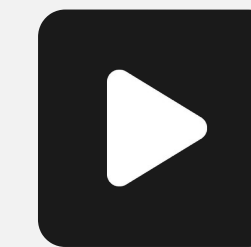
The governed sub-tier map shows which parts, products, and customers depend on that site, with confidence on each link.



Step 3

Decide

The part owner reviews the evidence, confirms exposure with tier 1, and approves an allocation plan.



Step 4

Act

Buyers start alternate-source qualification and account teams warn affected customers early.

The human decision stays in the loop. Governance makes it faster, not optional.

TAKEAWAYS

Three things you can start next quarter

1



Map your riskiest input

Pick one sole- or single-source material and trace it as far down the tiers as you can.

2



Give every critical record an owner

Name who keeps each critical part and supplier record accurate, and agree data-use terms with your key suppliers.

3



Put AI behind validation and a person

Test any risk model against past disruptions, and keep a named human decision point for actions it informs.

Thank you

See your risk sooner. Act on it with data you can defend.

Questions?

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